

GEOPHYSICS

Seismic evidence for partial melting at the root of major hot spot plumes

Kaiqing Yuan¹ and Barbara Romanowicz^{1,2,3*}

Ultralow-velocity zones are localized regions of extreme material properties detected seismologically at the base of Earth's mantle. Their nature and role in mantle dynamics are poorly understood. We used shear waves diffracted at the core-mantle boundary to illuminate the root of the Iceland plume from different directions. Through waveform modeling, we detected a large ultralow-velocity zone and constrained its shape to be axisymmetric to a very good first order. We thus attribute it to partial melting of a locally thickened, denser- and hotter-than-average layer, reflecting dynamics and elevated temperatures within the plume root. Such structures are few and far apart, and they may be characteristic of the roots of some of the broad mantle plumes tomographically imaged within the large low-shear-velocity provinces in the lower mantle.

The region of the mantle right above Earth's core is a boundary layer for mantle convection, and a chemical interface between the silicate mantle and the fluid iron core, where complex dynamic processes are thought to take place (1). The strong lateral variations in seismic structure in the bottom ~300 km of the mantle (the D'' region) are difficult to explain with temperature variations alone. Ultralow-velocity zones (ULVZs) are thin, localized patches (15 to 30 km in height) near the core-mantle boundary characterized by compressional and shear velocity reductions (5 to 10% and 10 to 30%, respectively) (2, 3), along with density increases (up to 10%) (4), with respect to ambient mantle. Most ULVZs reported are in regions of D'' with lower-than-average shear velocity, imaged by seismic shear-wave velocity tomography under the central Pacific and Africa (1, 5), and referred to as large low-shear-velocity provinces (LLSVPs). The origin of ULVZs

is thought to be either zones of partially molten silicates (6), possibly enriched in iron from the core (7), or dense iron-enriched solid heterogeneities (8, 9). The nature of ULVZs can clarify our understanding of the thermal and chemical structure near the core-mantle boundary. For example, some geodynamics simulations have modeled ULVZs as passive blobs of denser-than-average material. In this scenario, they would preferentially collect on the edges of the LLSVPs (10).

ULVZs may come in different sizes and shapes, although most previous studies have reported lateral extents of no more than 100 to 200 km (7). Determining their morphology is important for understanding their nature and dynamics. The shape of ULVZs is hard to constrain given available data, so estimates of their possible lateral extent are often based only on Fresnel zone considerations (11). This has led to the suggestion of a global correlation between the locations of ULVZs

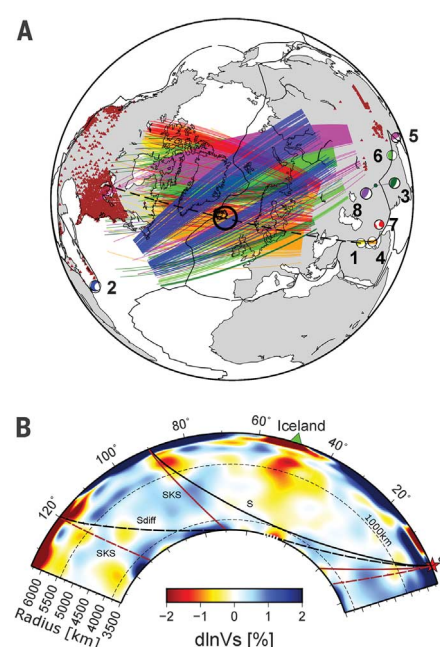
and hot spot volcanoes (11). However, this link and its causality have remained hypothetical.

Two examples of unusually large ULVZs (800 to 1000 km in lateral extent) were reported in the Pacific, one in the vicinity of the Samoa hot spot (12) and the other in the vicinity of Hawaii (13). Global tomographic images of the lower mantle obtained using numerical wavefield computations (14) showed that these ULVZs are located at the roots of broad quasi-vertical low-shear-velocity conduits that extend from the core-mantle boundary to at least 1000 km depth, in the vicinity of those hot spot volcanoes that fall within the boundaries of the LLSVPs (15). The precise shape of the Samoa ULVZ could not be constrained, but for the Hawaiian ULVZ, the azimuthal sampling by shear-diffracted (S_{diff}) phases, afforded by the USArray deployment in North America, showed that a simple cylindrical model is a good fit to the available data (13). However, illumination was limited to a particular corridor from the southwest Pacific to North America, making it difficult to distinguish between a circular base and one with a more elongated or irregular shape.

Notably, the Iceland plume was imaged tomographically as a broad low-velocity conduit extending down to the core-mantle boundary (15) (Fig. 1B). The presence of a ULVZ in this region was previously suggested from observations and modeling of $SPdKS$ and $SKPdS$ waves on a limited number of paths (16), not enough to constrain the lateral extent and shape of this structure. Although there was no tomographic evidence for a lower mantle plume under Iceland at that time, Helmberger *et al.* (16) suggested that this ULVZ could be a partially molten zone related to such a plume.

We used forward modeling of S_{diff} waves that sample the base of the mantle near the root of the Iceland hot spot, providing illumination from several distinctly different directions (Fig. 1, A and B). The S_{diff} phase spends a large portion of its path along the core-mantle boundary, where it behaves similarly to a surface wave. Waveform distortions, as a function of azimuth and distance, constrain both lateral and height variations of causative anomalous structures. Transverse component waveforms (SH and SH_{diff}), filtered in the period range 10 to 20 s and arranged as a function of azimuth in a narrow distance range for our main event (event 1 in table S1 and Fig. 2B), show secondary arrivals in the coda of the S_{diff} phase, at azimuths where the turning point of S waves and the diffracted portion of S_{diff} waves sample the vicinity of the root of the Iceland plume (Fig. 2A). The observed waveforms present several striking features. (i) A strong postcursor is delayed from the main S_{diff} wave arrival by about 25 s in the azimuth range -27° to -35° , with a maximum amplitude as large as that of the main phase away from this azimuth range. (ii) A strong reduction

Fig. 1. Geometry of the ULVZ study. (A) Map showing the illumination of the Iceland plume root region by seismic waves considered in this study. The numbers and "beach balls" correspond to earthquakes listed in table S1. S_{diff} paths sampling the lowermost 300 km of the mantle are color-coded according to event. Brown triangles indicate locations of recording stations. The black circle indicates the preferred location and size of the proposed axisymmetric ULVZ, plotted at the core-mantle boundary. Iceland and continents are plotted at Earth's surface. (B) Depth cross section through the Iceland plume in the model SEMUCB_WM1 (15), corresponding to the dashed line in (A), showing lateral variations in shear velocity (V_s) relative to the global mean ($d \ln V_s = dV_s/V_s$). The thin black dashed lines mark depths of 1000 and 2600 km. The broken line segment on the core-mantle boundary shows the preferred location and lateral extension of the ULVZ. The location of the Iceland hot spot is indicated by a green triangle; the red star indicates the event epicenter. S , S_{diff} (black), and SKS (brown) ray paths are indicated by solid and dashed lines, spanning the epicentral distance range available for event 1.



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in amplitude of the main phase is observed where the postcursor is largest. (iii) There is also a move-out of the postcursor, with fading amplitudes, at azimuths -35° to -42° . We verified that these effects originate at the base of the mantle by inspecting the corresponding *SKS* waveforms, which have similar paths as *S_{diff}* waves in the upper mantle (Fig. 1B) but do not exhibit the same waveform disturbances (fig. S1).

To constrain the causative structure, we used the same approach as developed earlier for the study of the Hawaiian ULVZ (13), applied to densely sampled broadband seismic waveform data (17). We forward-modeled synthetic waveforms in models that include three-dimensional (3D) structure in shear velocity in the bottom 400 km of the mantle (17). We used a coupled spectral-element/normal-mode code (18) previously applied to model structures in *D'* at the border of the African LLSVP (19) and under Hawaii (13). The approach that we used allowed a relatively fast calculation down to a cut-off period of 8 s. Reaching this period is essential for modeling the subtle waveform effects that we observed.

We could not reproduce the pattern observed in the waveforms of event 1 (Fig. 2B) by computing synthetics in tomographic models, because the

velocity heterogeneity is too smooth and too weak in these models (Fig. 2D). A taller, mushroom-like structure, but with less extreme low velocity than a ULVZ (20, 21), cannot explain this pattern, either (fig. S2E). To reproduce the observed waveforms to first order, we introduced a simple trial structure: a homogeneous, thin, cylindrical ULVZ with a vertical axis of symmetry, extending from the core-mantle boundary into the mantle, superimposed on the tomographic model SEMUCB_WM1 (15). We compared the 3D synthetics generated by varying the position, radius, and height of the cylinder and the internal velocity reduction with the corresponding waveform data for event 1. Our best-fitting models are located close to the Iceland hot spot, with a diameter of $\sim 800 \pm 50$ km, a height of $\sim 15 \pm 5$ km, and a shear velocity reduction of $\sim 23 \pm 5\%$ (e.g., Fig. 1B, figs. S3 and S4, and table S2). The corresponding 3D synthetics (Fig. 2C and fig. S5) reproduce the main characteristics of the observed waveforms well—in particular, the arrival time of the postcursor to within a few seconds.

We do not expect this model to provide a perfect fit to the observed waveforms, because mantle heterogeneities located higher than 400 km above the core-mantle boundary may introduce

waveform distortions, and the radiation pattern of the earthquake may slightly differ from the centroid moment tensor (CMT) solution (17) at these short periods. The sides of the structure are unlikely to be perfectly vertical. Trade-offs also exist between cylinder radius, velocity reduction, and height. In addition, there are a few degrees of uncertainty in the ULVZ position.

The data from event 1 by itself span a limited range of azimuths. Importantly, the same cylindrical model predicts the main characteristics of *S_{diff}* waveforms and their codas for seven other events that we did not use in its construction (Fig. 2, E to L, and figs. S6 to S12), for which the waveforms span a wide range of azimuths around the ULVZ (Fig. 1A) and a combined distance range of over 40° (table S1), thus illuminating the region from several different directions and angles. An axisymmetric ULVZ is expected to produce a postcursor with a hyperbolic shape move-out, symmetric with respect to the great circle linking the source to the center of the ULVZ (e.g., fig. S13). We further demonstrated that the base of the ULVZ under Iceland is close to circular in shape by producing composite record sections, combining waveforms from events 1 and 3 (Fig. 3, A to C). We did this by rotating the reference frame for one

Fig. 2. Comparison of *S* and *S_{diff}* waveform data and synthetics for events 1 to 3. Data and synthetics are shown for the transverse component (SH), filtered in the period range 10 to 20 s. (A to D) Event 1, in the distance range 94° to 96° . (A) Geometry of the illumination of the ULVZ structure, with the portion of the ray path sampling the last 300 km above the core-mantle boundary color-coded by azimuth to match (B) to (D). The pink circle indicates the location of the best-fitting cylindrical ULVZ. The red dashed line indicates the great circle passing through the epicenter and the center of the ULVZ. Green dots represent stations. (B) Observed waveforms. (C) 3D synthetics for the preferred ULVZ model. (D) 3D synthetics computed in the tomographic model SEMUCB_WM1 (14). Amplitudes of synthetics have been normalized to those of the data at azimuths far away from the ULVZ to account for adjustments needed to the source CMT solution (www.globalcmt.org). Waveform distortions in (A) between azimuths -25° and -35° are well explained by the model in (C) and completely absent in the synthetics in (D). Other examples of fits for event 1 are shown in fig. S5. (E to H) Same as (A) to (D), but for event 2, observed in the distance range 126° to 130° . (I to L) Same as (A) to (D), but for event 3, observed in the distance range 101° to 105° . The 3D synthetics in (G) and (K) are computed for the preferred model obtained for event 1. This model has not been adjusted to fit the waveform data for these two events. The data shown for events 2 and 3 sample not only different azimuths, but also different distance ranges than those shown for event 1. The increasing travel-time delay between azimuths -23° and -28° for event 3 may be attributed to the passage of the direct *S_{diff}* phase through a low-velocity region that is visible under Scandinavia in SEMUCB_WM1 (e.g., fig. S1A), but not fully resolved in that model.

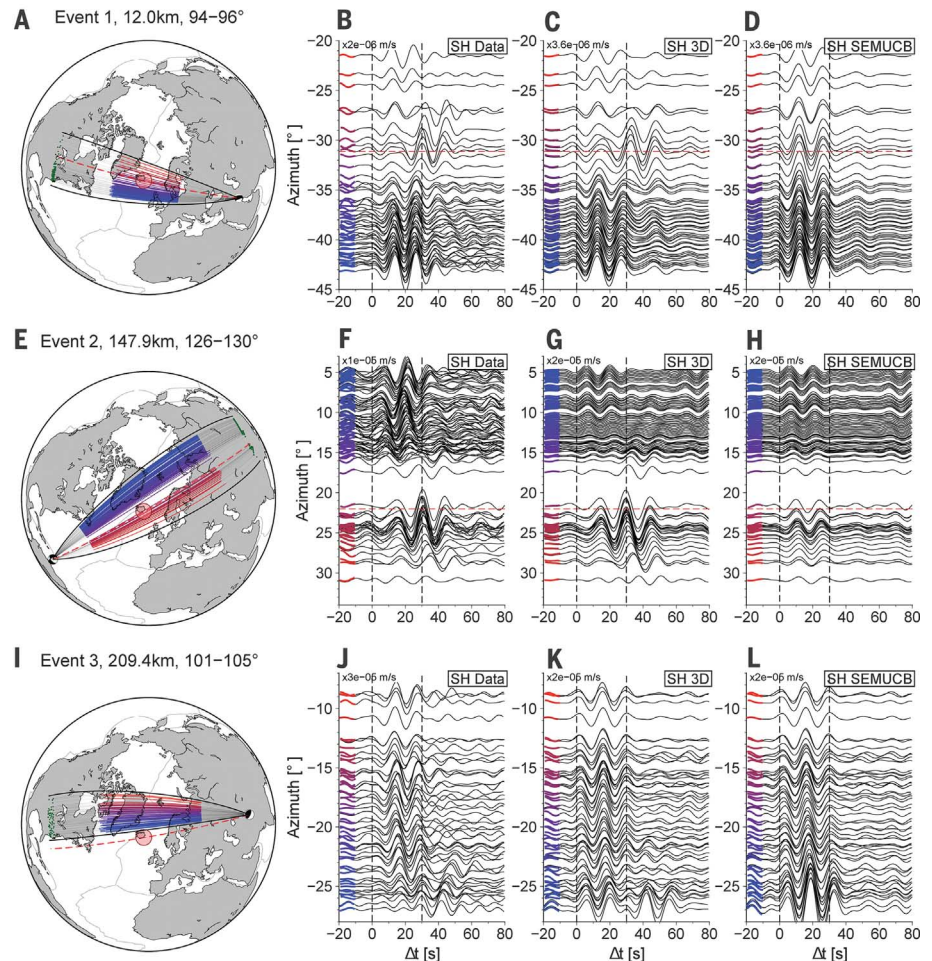


Fig. 3. Combined record sections for event pairs. (A to C) Event pair 1 and 3. The record sections for both events have been combined, after applying the rotation that brings event 3 (red) in line with event 1 (black). (A) Map showing the azimuth range and distance range of stations after rotation. The red arrow shows the rotation applied. The true location of event 3 is also shown. Dashed lines indicate the azimuth range shown in (B) and (C), the solid line is the great circle that defines azimuth 0°, and the red circle shows the position of the ULVZ. (B) Combined transverse component record section of the data, filtered between 10 and 20 s. (C) Combined record section for the corresponding synthetics, assuming our preferred cylindrical ULVZ model. (D to F) Same as (A) to (C), but for event pair 2 and 3. In all record sections, a theoretical postcursor diffraction hyperbola and the trend of travel-time delay in the main S and S_{diff} phases, assuming perfect cylindrical symmetry, are indicated for reference (blue shading). Slight mismatches in waveforms and arrival times can be explained by a combination of imperfect radiation pattern and heterogeneities outside of the region considered. The difference in distance ranges for event pairs also contributes to mismatches, as can be seen in the synthetics computed for a perfectly cylindrical ULVZ.

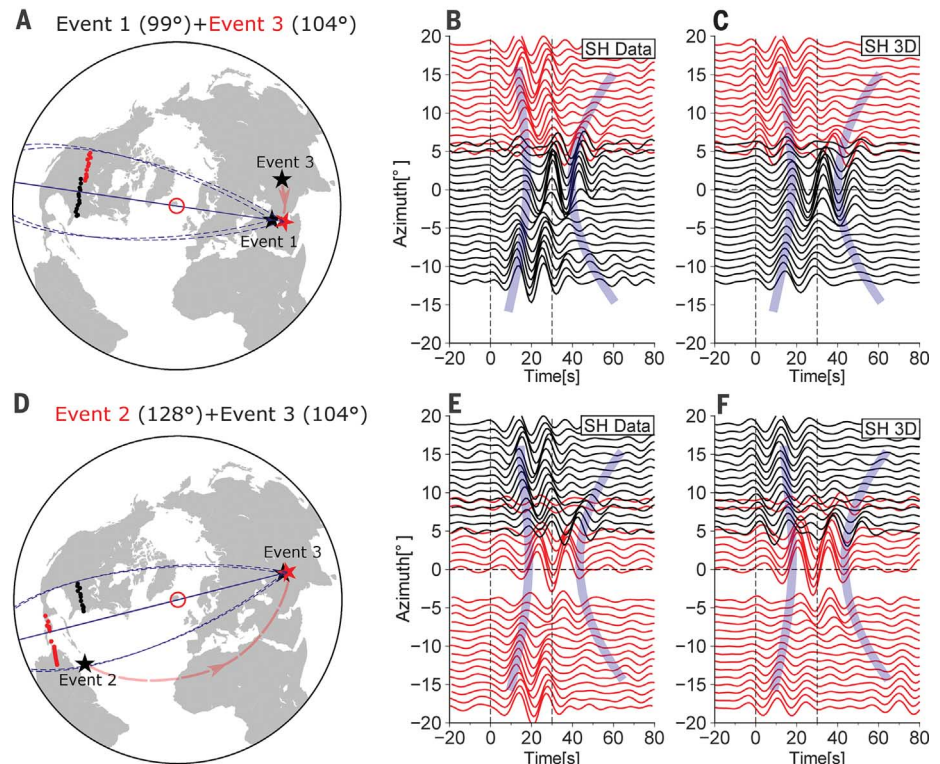
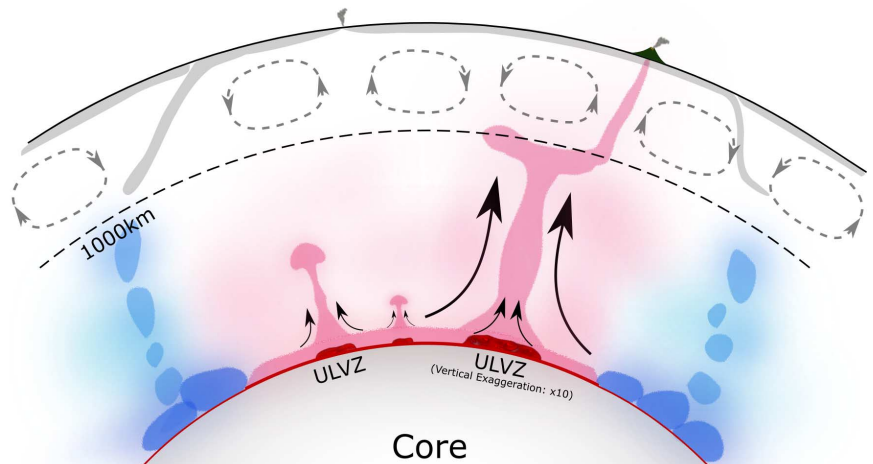


Fig. 4. Conceptual sketch of ULVZs (red) as topographic hills in a thin, dense, partially molten layer at the core-mantle boundary. The ULVZs are vertically exaggerated by a factor of 10 relative to the thickness of the mantle and the height of the plumes (pink). A large, axisymmetric ULVZ resides under the broad mantle plume, whereas smaller ones may exist beneath less fully developed upwellings. The thick dashed line indicates the depth of 1000 km where many large plumes are deflected horizontally (15) and some slabs (gray) stagnate (41). Sinking slab fragments are shown in blue. Arrows indicate the direction of flow. The LLSVPs may not be coherent structures across the depth of the lower mantle, but rather may represent plume clusters (39), still imperfectly resolved by tomography, which results in a background long-wavelength low-velocity halo.



event with respect to the other until the source locations illuminated the ULVZ from the same angle. This allowed us to reconstruct a more complete arc of the diffraction hyperbola, which is only possible if the ULVZ is close to axisymmetric. Such a rotation also works well for the pair of events 2 and 3, noting that event 2, in its original location, illuminates the ULVZ from a completely different side (Fig. 3, D to F).

In addition, the S_{diff} waveforms are measurably affected only for paths that approach the modeled ULVZ to within a distance that is less than ~800 km—i.e., one diameter away from the structure—indicating that this large ULVZ is the

sole feature of its kind in a broad region of D'' along the Mid-Atlantic Ridge on either side of Iceland. Indeed, for azimuths further away from the ULVZ (e.g., paths marked in blue in fig. S6A, and all paths in fig. S6, B and C), which sample D'' as far as five times the diameter of the ULVZ to the north of it (fig. S6B) and two times to the south of it (fig. S6A), we found no evidence for postcursors that could be associated with another similar ULVZ.

The quasi-axisymmetric shape of this ULVZ, its location at the root of the Iceland plume as imaged tomographically (15), and its large diameter, commensurate with the width of the plume

higher up in the lower mantle (15), imply a close dynamical link between the ULVZ and the plume. This suggests that the ULVZ most likely involves partial melting (6, 22, 23). The ULVZ would be confined to the hot base of the plume (24, 25), and its borders may mark the transition between a partially molten zone and the solid surroundings. A compositionally distinct, solid ULVZ of low-viscosity material (26, 27) could possibly be shaped similarly by plume dynamics; however, this would necessitate explaining how a heterogeneous solid blob of the right size could be swept right into the plume root and not elsewhere. The shear velocity reduction of 23%, as argued in previous

studies (6, 28), is compatible with a melt fraction of 10 to 20% by volume, although our data cannot provide further constraints, because they are insensitive to P velocity reduction and density. Although our approach cannot directly constrain the density within the ULVZ, it has been argued that melt in D'' would be denser than the surrounding solid (6), and the stability of hot spots through substantial geological time requires the presence of denser-than-average material at the roots of the corresponding mantle plumes (24, 25, 29).

The large ULVZ that we imaged may not be representative of all ULVZs (30). However, its unique occurrence in an extended region of the core-mantle boundary, combined with previous observations and modeling of a similar ULVZ at the base of the Hawaiian plume (13) and a “mega” ULVZ beneath Samoa (12), is striking. This suggests the existence of a specific class of large ULVZs formed at the roots of broad plumes associated with major, presently active hot spots (15). Such ULVZs may be the clearest manifestation of thickening of an otherwise likely very thin (kilometer-scale), partially molten layer (31, 32) (Fig. 4) that is enriched in iron and therefore denser than its surroundings. This layer may represent an interaction zone between mantle and core (7), which could be the source of isotopic anomalies of ancient origin observed at this and other hot spots (33, 34). Evidence for such a thin layer has been suggested from fine-scale local seismic studies in the Pacific (31) and elsewhere (32). Other ULVZs not associated with the broad plumes imaged tomographically may also mark local hills in this layer caused by smaller, hot upwelling instabilities in the LLSVPs (Fig. 4). The denser-than-average part of the LLSVPs may be limited in height to the locations of these ULVZs, which may explain why recent travel-time and normal-mode studies that do not have much sensitivity to the last 100 km at the base of the mantle can explain the long-wavelength structure of the LLSVPs by lateral variations in temperature alone (35, 36). This poses the question of how exactly one should define the height of a LLSVP. LLSVPs are often described as large piles of compositionally distinct material (12, 37, 38), but they may actually be the long-wavelength expression

of a bundle of plumes (15, 39), anchored in a thin dense layer with high topography at its top (Fig. 4).

Our results should provide guidance for further modeling of the thermal history, mineralogy, and present-day dynamics at the very base of the mantle. Further characterizing the fine-scale structure of the bottom 50 km of the mantle—and, in particular, detection of ULVZs at the root of other broad mantle plumes—would be facilitated by the deployment of new large-aperture broadband arrays, especially on the ocean floor, to increase illumination of the vicinity of the core-mantle boundary. This needs to be combined with the development of waveform imaging techniques that can target specific objects of limited extent at short enough periods, with realistic computational resources (40).

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SUPPLEMENTARY MATERIALS

www.sciencemag.org/content/357/6349/393/suppl/DC1
Materials and Methods
Figs. S1 to S13
Tables S1 and S2
References (42–45)

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Seismic evidence for partial melting at the root of major hot spot plumes

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Iceland's molten roots

Major hot spot plumes are responsible for basaltic ocean island chains such as Hawaii. Yuan and Romanowicz used seismic tomography, which constructs an x-ray–like picture of Earth's interior from seismic waves, to show that the root of Iceland's hot spot plume is partially molten. The partially molten region is located near Earth's core-mantle boundary and has been challenging to image with geophysical methods. This approach may be applicable to other hot spots with similar areas of melts or other enigmatic regions in the lower mantle.

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